TELIPORT
Script Your Future Season 4

TATA
TATA ELXSI

CASE STUDY 1 / CONNECTED FLEET PREDICTIVE MAINTENANCE

FlairGuard AI — A Verified Connected-Fleet Reliability Platform

Causal RUL prediction, telemetry-fault recovery and auditable maintenance decisions under constrained service capacity.

Team: _____ Members: _____ College: _____ Team ID: _____

NASA C-MAPSS → REGIME-AWARE PROGNOSTICS → TELEMETRY QUALITY & RECOVERY → DETERMINISTIC MAINTENANCE DECISIONS → PERSISTENT WORK ORDERS & AUDIT

NASA C-MAPSS
21 sensors, 3 settings
Past-only rolling windows

→

Regime Prognostics
LightGBM / HGB / Ridge
Regime-aware normalization

→

Quality & Recovery
Missingness & staleness
Abstention & 6 recovery methods

→

Deterministic Decisions
RUL & quality thresholds
Human review for exceptions

→

Work Orders & Audit
Priority, deduplication
SQLite audit & recovery

FD001/FD002 operational serving; FD003/FD004 retrospective research coverage · MATLAB toolbox comparison integrated

TRUST

Engine-disjoint training and causal inference; decision-horizon risk evaluation; six telemetry-recovery methods tested across eight controlled conditions.

VALUE

FD002 trigger-outage scenario: 0/80 modeled failures with regime imputation versus 14/80 with bounded countdown.

Scenario-specific retrospective result; no universal superiority claim.

EXECUTION

Working FastAPI + React operator platform, authenticated work-order workflows, restart-safe replay, model-version pinning, real MLflow training and rollback verification.

Real Chromium browser workflow and Docker restart checks completed.

114 passing tests · 332 primary policy runs · 17,822 scored engine episodes from 169 distinct source engines · Independent event-ledger verification · MATLAB toolbox comparison integrated

NASA C-MAPSS retrospective simulation; costs are hypothetical. Scientific model promotion requires fresh acceptance data.